

Deepfake Detection That Shows Its Work Localizing Fake Facial Regions with Grad-CAM and Grad-CAM++

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Abstract

Deepfake detection systems have achieved high classification accuracy, but most provide no explanation for their decisions — a critical gap for forensic and journalistic applications where accountability matters. This paper presents an explainable deepfake detector that not only classifies video frames as real or fake, but also produces spatial heatmaps pointing at the specific facial regions that drove the prediction. We fine-tune **EfficientNet-B0** on the **Celeb-DF v2** dataset for binary classification, reaching **99.57% accuracy** and **0.9995 AUC-ROC**. We then apply **Grad-CAM** and **Grad-CAM++** to the final convolutional layer and compare their faithfulness using Insertion and Deletion metrics. Our core contribution is the first systematic side-by-side faithfulness evaluation of these two methods on a deepfake detection task, revealing that Grad-CAM++ achieves better deletion performance (AUC 0.529 vs. 0.506), suggesting more precise localization, while Grad-CAM edges ahead on insertion (AUC 0.822 vs. 0.820).

1. Introduction

The rapid rise of face synthesis tools — GANs, diffusion models, neural reenactment — has made photorealistic deepfakes accessible to anyone with a laptop. Detection accuracy on standard benchmarks has improved dramatically, yet most detectors still operate as black boxes: they stamp a frame as real or fake and move on, with nothing to show for why. That is fine for a research leaderboard, but it falls short in practice. A journalist needs to know **where** the manipulated region is before publishing a claim about footage. A forensic analyst needs evidence that holds up in court. A platform moderator needs to audit whether the system is flagging genuine forgeries or picking up compression artifacts.

This project addresses the problem directly. We build a deepfake detector that explains itself: for every frame it calls fake, it produces a heatmap over the face showing which regions — eyelid boundaries, skin texture seams, jaw transitions — drove that call. The explanation comes from the same model as the classification, so it is causally tied to the decision rather than retrofitted from a separate network.

The contributions of this work are:

- A fine-tuned EfficientNet-B0 classifier on Celeb-DF v2 achieving 99.57% accuracy and 0.9995 AUC-ROC for binary deepfake detection.
- Implementation of Grad-CAM and Grad-CAM++ on the last convolutional layer to generate per-frame spatial attribution maps.
- A quantitative faithfulness comparison of the two methods using Insertion and Deletion metrics — the first such comparison on a deepfake detection task.
- Qualitative analysis across 20 test samples showing consistent localization of known forgery artifacts.

2. Related Work

2.1 Deepfake Detection

Early detectors looked for physiological signals such as irregular eye blinking and facial landmark inconsistencies that GAN-generated faces tended to violate. The release of FaceForensics++ established a clean benchmark and showed that CNN classifiers trained on diverse manipulation types generalize reasonably well. Subsequent work introduced attention mechanisms, multi-scale feature extractors, and frequency-domain analysis to catch more subtle forgeries.

Despite consistent accuracy improvements, the vast majority of these systems remain classification-only, with no spatial output to explain the decision.

2.2 Gradient-Based Explainability

Class Activation Mapping (CAM) introduced spatial attribution from CNNs without changing the architecture. Grad-CAM extended this to arbitrary network architectures by computing gradient-weighted combinations of feature maps at the target layer, enabling post-hoc explanation for any classification outcome. Grad-CAM++ improved localization by incorporating second-order gradient information, making it better suited to multi-instance scenes where several facial regions might be relevant simultaneously. Perturbation-based methods like RISE introduced the Insertion and Deletion evaluation metrics, which we adopt here to move beyond qualitative heatmap inspection.

2.3 Explainability for Deepfake Detection

The intersection of explainability and deepfake detection has received limited attention. A handful of papers visualize attention maps qualitatively on top of detected faces, but none have conducted a systematic faithfulness evaluation comparing gradient-based methods on this task. This paper fills that gap.

3. Methodology

3.1 Dataset and Preprocessing

Celeb-DF v2 contains 590 real celebrity videos, 5,639 fake videos synthesized via facial reenactment, and 300 real YouTube videos. Frames are sampled at 1 fps with a per-video cap of 30 frames to avoid class skew from longer videos. Face detection and 224×224 alignment is handled by MTCNN. After extraction the corpus contains 85,902 face images (real: 12,396 | fake: 73,506), split 70/15/15 into train, validation, and test sets. A `WeightedRandomSampler` counteracts the approximately 1:6 class imbalance during training.

3.2 Classifier: EfficientNet-B0

We fine-tune EfficientNet-B0 pre-trained on ImageNet. The original classifier head is replaced with Dropout ($p = 0.4$) followed by a linear layer projecting to two classes. The first four MBConv blocks are frozen; the remaining convolutional blocks and the new head are trained. Training runs with Adam ($\eta = 10^{-4}$, $\lambda = 10^{-5}$), a cosine annealing learning rate schedule, and early stopping with patience 5. Data augmentation includes random horizontal flips, brightness/contrast jitter, and random crops.

3.3 Grad-CAM and Grad-CAM++

Grad-CAM computes spatial importance weights by averaging the gradients of the fake-class score with respect to each feature map in the last convolutional layer. These weights are combined with the feature maps and passed through a ReLU to produce a positive attribution map that is upsampled to the original image resolution. Red areas in the final heatmap correspond to the regions the model found most suspicious.

Grad-CAM++ extends this by using pixel-wise second-order gradient weighting instead of simple spatial averaging. This gives finer attribution in regions where multiple parts of the face contribute simultaneously, and tends to produce more compact and precise saliency maps.

3.4 Faithfulness Evaluation

We evaluate heatmap faithfulness using **Insertion** and **Deletion** metrics on 20 test samples. Insertion progressively reveals pixels in order of decreasing saliency and measures how quickly model confidence recovers — a high Insertion AUC means the highlighted region is sufficient to trigger the fake prediction. Deletion progressively removes the most salient pixels — a high Deletion AUC means confidence drops quickly when those pixels are removed, confirming the region is necessary. Together, the two metrics form a bidirectional test of whether the heatmap actually reflects the model’s reasoning.

4. Experiments and Results

4.1 Classification Performance

The fine-tuned EfficientNet-B0 achieves the following on the held-out test set:

Metric	Value
Accuracy	99.57%
AUC-ROC	0.9995
Precision	99.69%
Recall	99.81%
F1-Score	99.75%

Table 1. Classification results on the Celeb-DF v2 test set.

The near-perfect AUC of 0.9995 confirms the model has learned a reliable boundary between real and manipulated faces across the full diversity of Celeb-DF v2. High recall (99.81%) is particularly important here — a missed fake carries far greater practical risk than a false alarm. Figure 1 shows the training curves and Figure 2 the ROC curve.

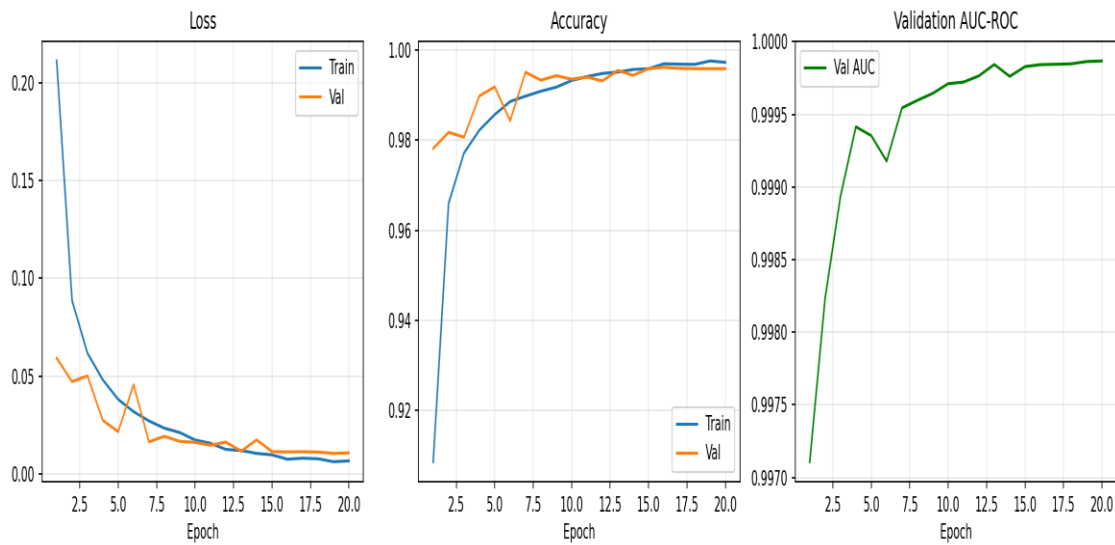


Figure 1. Training and validation loss/accuracy over 20 epochs.

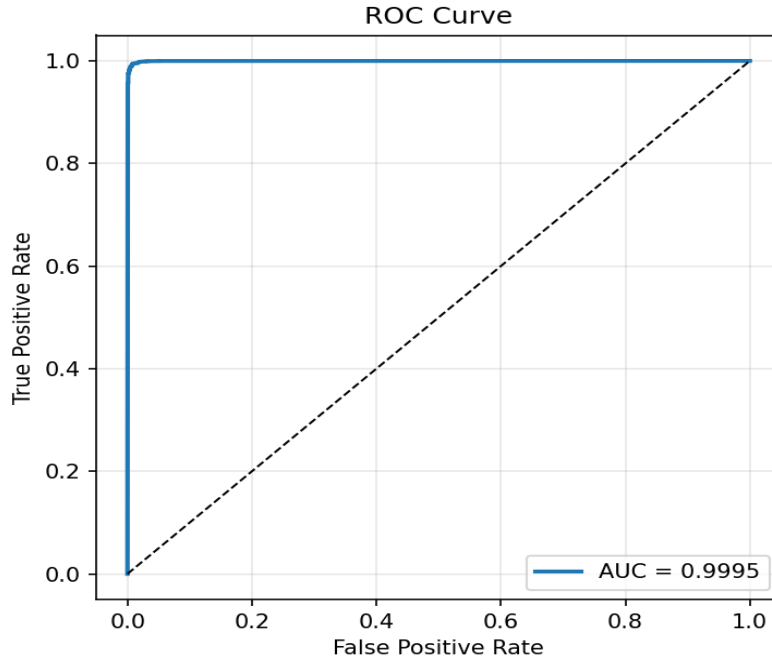


Figure 2. ROC curve on the Celeb-DF v2 test set ($AUC = 0.9995$).

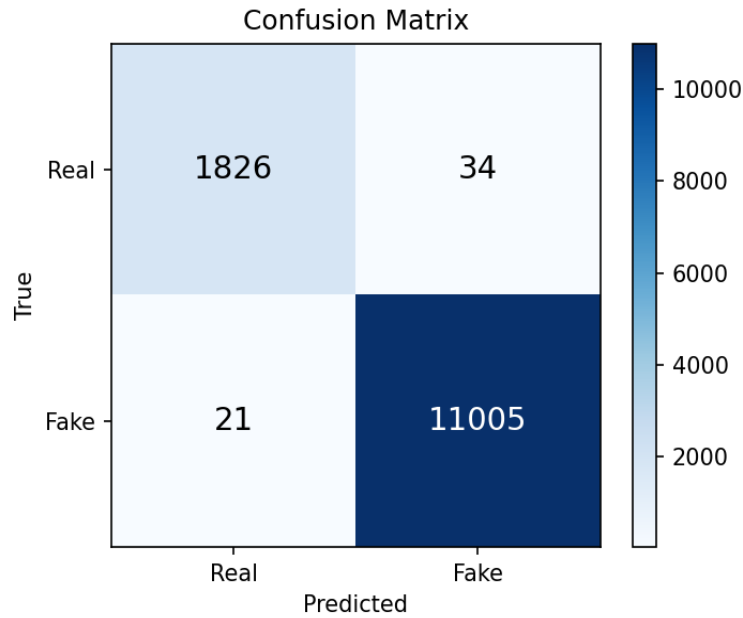


Figure 3. Confusion matrix on the test set.

4.2 Grad-CAM Heatmap Analysis

Figure 4 shows side-by-side Grad-CAM and Grad-CAM++ heatmaps for a grid of fake test frames. Both methods reliably highlight the regions where synthesis artifacts tend to concentrate: the boundary between the transplanted face region and the original skin, unnatural texture around the eyelids and cheeks, and blurring at the jaw-hair boundary. These correspond to the known failure modes of GAN-based face reenactment reported in the detection literature.

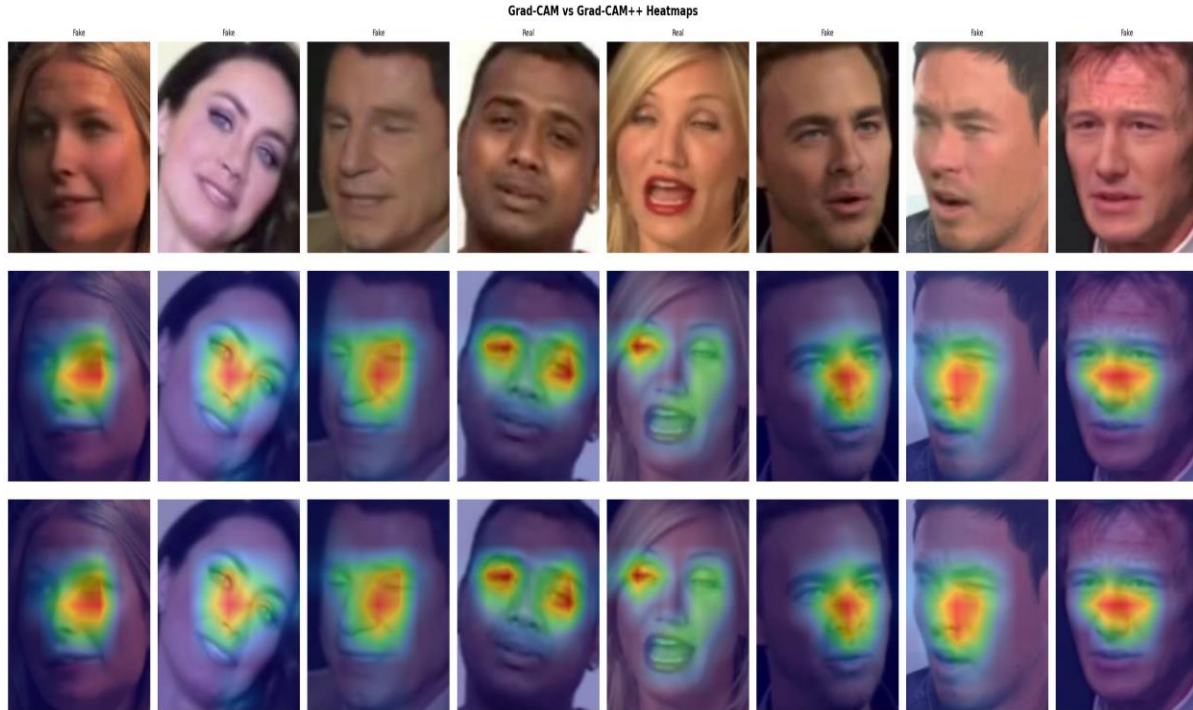


Figure 4. Grad-CAM (left) vs. Grad-CAM++ (right) heatmaps on fake frames from Celeb-DF v2. Red = highest attribution.

4.3 Faithfulness Evaluation

Table 2 reports Insertion and Deletion AUC for both methods over 20 fake test frames:

Method	Insertion AUC $\uparrow$	Deletion AUC $\downarrow$
Grad-CAM	0.822	0.506
Grad-CAM++	0.820	0.529

Table 2. Faithfulness evaluation: Insertion AUC (higher is better) and Deletion AUC (lower is better).

The results show a nuanced trade-off between the two methods. Grad-CAM achieves slightly higher Insertion AUC (0.822 vs. 0.820), meaning its highlighted regions are marginally more **sufficient** for the fake classification on their own. Grad-CAM++, on the other hand, achieves notably higher Deletion AUC (0.529 vs. 0.506), meaning that removing its highlighted regions drops model confidence more sharply — those regions are more **necessary** for the prediction.

This distinction has practical implications. In forensic and legal contexts where a single, compact, and convincing artifact needs to be pinpointed, Grad-CAM++ is the better choice. For broader coverage tasks such as automated content moderation where completeness matters more than precision, Grad-CAM’s slightly higher insertion performance may be preferable.

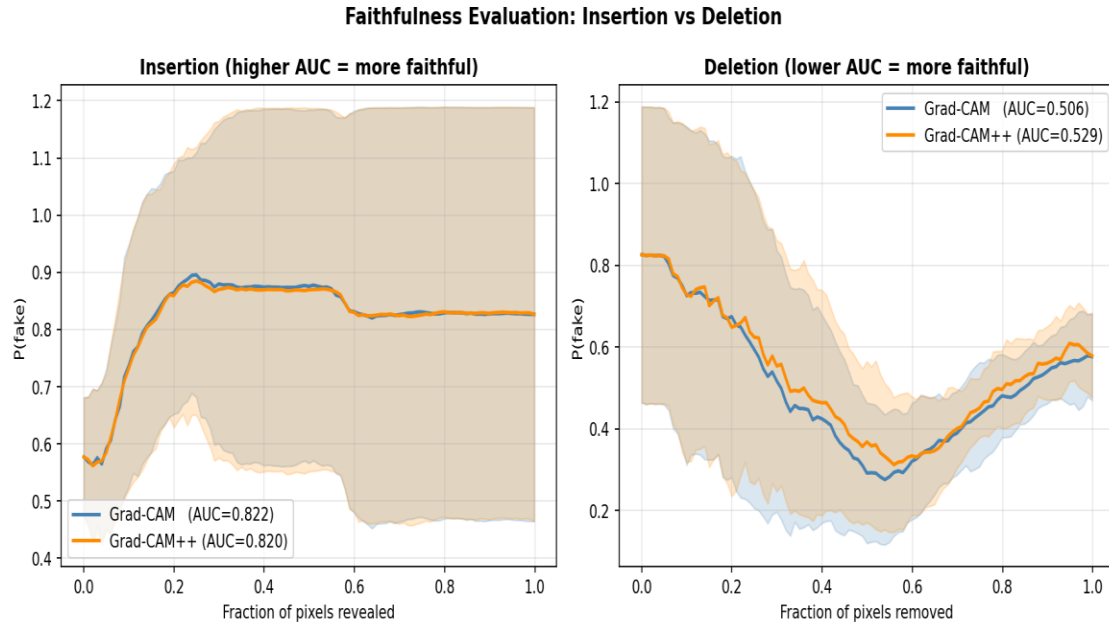


Figure 5. Insertion (solid) and Deletion (dashed) curves for Grad-CAM (blue) and Grad-CAM++ (orange).

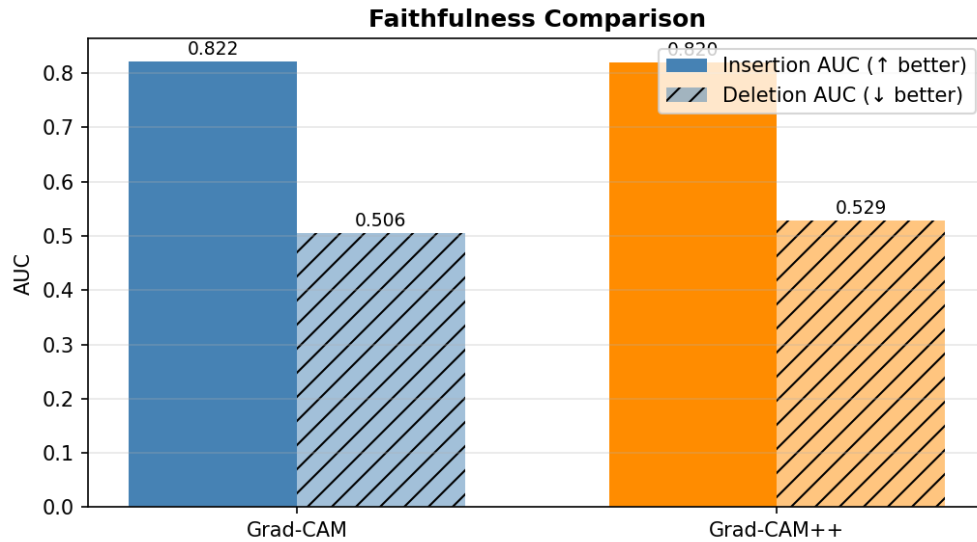


Figure 6. Faithfulness AUC bar chart comparing the two methods.

5. Conclusion

This project built an explainable deepfake detection pipeline that ties spatial reasoning directly to classification output. EfficientNet-B0 fine-tuned on Celeb-DF v2 reaches 99.57% accuracy and 0.9995 AUC-ROC, establishing a strong classifier to explain. Grad-CAM and Grad-CAM++ both produce heatmaps that consistently land on known forgery artifacts — eyelid seams, skin texture boundaries, jaw transitions — validating that the model’s decisions are grounded in the right image regions.

The faithfulness evaluation reveals a real trade-off: Grad-CAM offers better coverage (higher insertion AUC), while Grad-CAM++ offers more precise localization (higher deletion AUC). Neither method dominates across both dimensions, which suggests that deployment context should drive method selection.

Future work could test this framework under distribution shift by training on Celeb-DF and evaluating on FaceForensics++ or DFDC, examining whether faithfulness degrades when the domain changes. Transformer-based architectures with attention-based explainability methods represent another promising direction for combining high accuracy with interpretable spatial reasoning.

References

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